

RESEARCH ARTICLE

YOLOv13-Based Robust Underwater Object Detection via Illumination-Invariant Deformable Alignment

GAOQUAN GU^{ID}, KAI LIU, AND QINGCHAO YANG

Naval University of Engineering, Wuhan 430033, China

Corresponding authors: Kai Liu (0905061015@nue.edu.cn) and Qingchao Yang (1405061065@nue.edu.cn)

ABSTRACT To address the problems of false detections, missed detections, and unstable localization in underwater target detection under non-uniform illumination, turbidity-induced scattering, and occlusion, this paper proposes a robust underwater object detection method based on YOLOv13. The proposed method is designed around two key bottlenecks: discriminative reliability under non-uniform illumination and localization stability under occlusion and overlap. First, a learnable Illumination Invariant Module (IIM) is constructed based on illumination-invariant color-ratio theory to suppress brightness-field interference through zero-mean constrained convolution and feature fusion. Second, Bi-level Routing Attention (BRA) is incorporated into the high-level feature-fusion stage to selectively aggregate reliable contextual information and reduce interference from pseudo-textures and background highlights. Third, Deformable Convolution (DCN) is introduced into the key prediction layers to improve localization robustness in occluded and overlapping scenes through adaptive sampling and structural alignment. Experimental results show that the proposed method achieves an mAP of 95.24% at 115 FPS on the UT dataset and 84.91% at 113 FPS on the URPC2020 dataset, outperforming the YOLOv13 baseline by 2.31 and 0.76 percentage points, respectively, while maintaining favorable real-time performance. Ablation studies further verify the effectiveness of the coordinated integration of IIM, BRA, and DCN across the feature representation, feature-fusion, and prediction stages. These results demonstrate that the proposed method can provide more stable and accurate underwater target detection under complex imaging conditions.

INDEX TERMS Underwater object detection, illumination-invariant features, bi-level routing attention, deformable convolution, YOLOv13.

I. INTRODUCTION

Rapid and accurate detection of hazardous targets such as divers and underwater vehicles is therefore a key prerequisite for achieving early detection and timely decision-making in informationized maritime operations [1]. In this context, underwater vehicles, owing to their high degree of autonomy and long-endurance mission capability, are increasingly emerging as one of the core platforms for underwater reconnaissance, surveillance, information acquisition, and target search [2], [3]. However, the optical sensing systems

carried by Underwater vehicles are inevitably affected in real underwater environments by factors such as absorption and scattering, turbidity-induced degradation, and variations in illumination conditions, which lead to reduced image contrast and amplified noise, thereby substantially increasing the difficulty of underwater target detection [4], [5]. This challenge is particularly pronounced in deep-water scenarios, where non-uniform illumination often causes local overexposure, underexposure, and shadow artifacts, resulting in strong spatial inconsistency in target texture visibility and boundary saliency, and further giving rise to false detections, missed detections, and unstable localization.

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Early image-based target detection methods generally followed a two-stage pipeline of “enhancement first, detection second.” In the preprocessing stage, researchers commonly employed histogram equalization, contrast stretching, white balancing, and color correction to improve low-contrast and color-distorted appearances, often in combination with Retinex decomposition and dehazing/descattering strategies to enhance edge and texture details, thereby providing more stable inputs for subsequent detection [6], [7], [8]. In the detection stage, these methods largely relied on handcrafted features such as edges, textures, and shape contours (e.g., gradients, LBP, and HOG), together with threshold segmentation, template matching, or conventional classifiers for target discrimination and localization. Although this serial “enhancement–feature–decision” paradigm can be effective under controlled conditions, it is highly sensitive to variations in imaging conditions. When local overexposure/underexposure, shadows, or noise amplification are present, enhancement strategies often struggle to balance detail preservation with noise suppression, which in turn leads to representational statistical drift and ultimately manifests as increased false positives, missed detections, and degraded localization stability.

With the development of deep learning, end-to-end detectors have gradually replaced the traditional serial pipeline of image enhancement, handcrafted feature extraction, and independent classification. Existing studies have improved detection performance through feature enhancement, occlusion-robust learning, lightweight design, and global-context modeling [9], [10], [11], [12]. Lei et al. [13] employed a Swin Transformer backbone and enhanced multi-scale feature fusion to strengthen the representation of targets at different scales. However, such methods generally assume that the input features are sufficiently reliable, whereas background highlights and scattering-induced pseudo-textures may also be enhanced and propagated under non-uniform underwater illumination. Zeng et al. [14] introduced adversarial occlusion training to improve the detection of partially occluded targets, but this strategy does not explicitly address the simultaneous degradation of target appearance caused by illumination variation and turbidity. Yang et al. [15] developed a lightweight detection network to reduce information loss and parameter redundancy, although its fixed local sampling operations remain sensitive to blurred and incomplete target boundaries. Lu et al. [16] enhanced low-level edge and corner information through multi-scale feature representation, while Bai et al. [17] introduced self-attention guidance and global feature fusion to model long-range dependencies. Nevertheless, these approaches may indiscriminately aggregate both reliable target cues and illumination-induced interference. Zhou et al. [18] improved real-time detection through dynamic label assignment and self-attention, but the method was mainly designed for general natural scenes and lacks targeted modeling of underwater illumination degradation and structural misalignment. Therefore, directly transferring these general detection

methods to underwater optical scenes cannot simultaneously guarantee reliable feature discrimination and stable target localization.

The above studies have advanced object detection algorithms in terms of backbone modeling capability, multi-scale feature fusion, and occlusion-robust training, substantially improving their adaptability to typical challenges such as dense targets and partial occlusion in general natural scenes. However, the performance gains of these methods are largely built on the premise that image observations are relatively clear and their statistical distributions remain stable. Once the imaging process is degraded by color shift, contrast reduction, scattering-induced haze, and non-uniform illumination, target textures and boundaries become spatially inconsistent, making the models more susceptible to pseudo-textures and background highlights while weakening the aggregation of effective structural information. This ultimately leads to simultaneous increases in false detections, missed detections, and unstable localization [19], [20], [21]. In underwater optical scenes, these factors are often coupled and superimposed. This issue is particularly pronounced in AUV cruising and search tasks with artificial illumination, where non-uniform lighting and occlusion coexist, making it difficult for general detection frameworks to achieve stable and reliable performance through direct transfer.

In recent years, many studies have addressed imaging degradation in underwater object detection. Dai et al. [22] proposed GCC-Net to fuse raw and enhanced image features, but its performance still depends on enhancement quality. Cen et al. [23] combined image enhancement with a lightweight YOLOv9 architecture, although preprocessing may amplify noise and background highlights under non-uniform illumination. Gao et al. [24] strengthened multi-scale and semantic feature interaction, but degradation-induced interference may also be propagated. Chen et al. [25] enhanced small-target representation through high-resolution multi-scale features, yet reliable target structures are not explicitly separated from pseudo-textures. Wang et al. [26] jointly optimized image processing and detection, but grayscale and edge enhancement may weaken useful color cues or become unstable under blurred boundaries. Li et al. [27] improved temporal feature modeling and efficiency using a spiking neural network, but did not directly address illumination-induced feature contamination or localization drift.

In summary, existing methods still face three limitations. First, non-uniform illumination may contaminate low-level representations and cause detectors to rely on unstable highlights or pseudo-textures. Second, conventional feature aggregation cannot effectively distinguish reliable target context from scattering and background interference. Third, fixed-grid convolution is sensitive to blurred, occluded, and overlapping boundaries, leading to localization drift. To address these issues, this paper develops a stage-wise collaborative detection method based on YOLOv13.

Specifically, at the representation stage, illumination-invariant feature extraction is constructed under imaging-physics constraints to reduce the interference of non-uniform illumination with low-level features. At the feature-fusion stage, routing-based sparse attention is incorporated to suppress pseudo-textures and background interference while strengthening the aggregation of reliable contextual information. At the prediction stage, deformable convolution is employed to achieve adaptive sampling and structural alignment, thereby improving localization stability under occlusion and overlap. Accordingly, the proposed method alleviates illumination interference, spurious-feature propagation, and localization drift from the perspectives of feature representation, contextual discrimination, and target localization, respectively, thereby improving detection accuracy and robustness in complex underwater scenes.

II. THEORETICAL BACKGROUND

A. THE PROPOSED METHOD FRAMEWORK

The proposed method is built upon YOLOv13 and retains its Hypergraph-based Adaptive Correlation Enhancement (HyperACE) and Full-Pipeline Aggregation-and-Distribution (FullPAD) mechanisms. On this basis, IIM is introduced

before the backbone to improve the stability of low-level representations under non-uniform illumination; BRA is incorporated into the high-level feature-fusion stage to enhance the aggregation of reliable contextual information; and selected depthwise separable convolutions in the prediction layers are replaced with deformable convolutions to improve adaptive sampling and structural alignment. Through the coordinated enhancement of feature representation, contextual discrimination, and target localization, the proposed framework improves detection robustness in complex underwater scenes. YOLOv13 enhances cross-scale feature interaction and full-pipeline representation coordination through HyperACE and FullPAD, providing a suitable baseline for the proposed framework [28]. The backbone first extracts multi-scale feature maps. Three selected feature maps are then aggregated and fed into HyperACE for cross-scale and cross-position correlation modeling and feature enhancement. The enhanced features are subsequently distributed by FullPAD to key locations, including the backbone–neck connection, the internal layers of the neck, and the neck–head interface. Finally, the multi-scale detection heads perform object classification and bounding-box regression. The overall architecture of the proposed method is shown in Figs. 1 and 2.

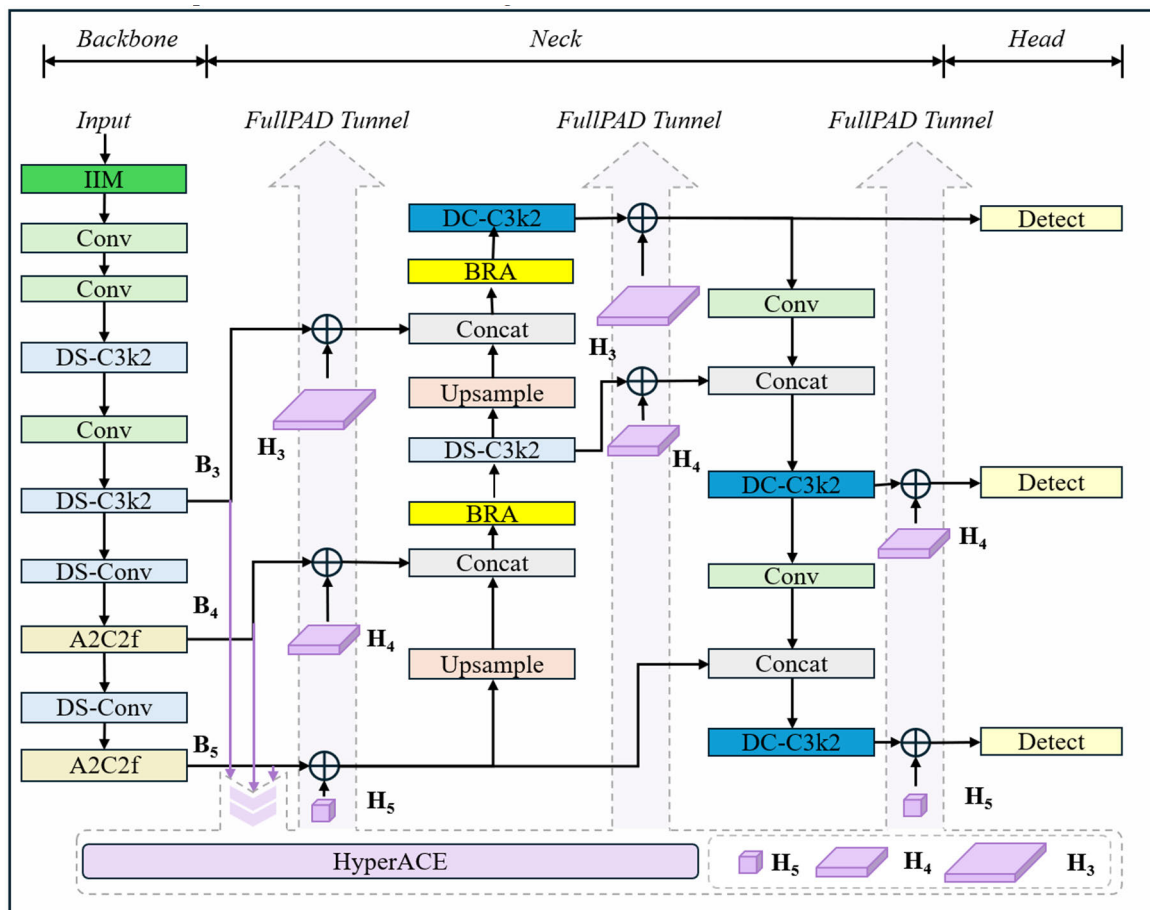


FIGURE 1. The proposed method architecture.

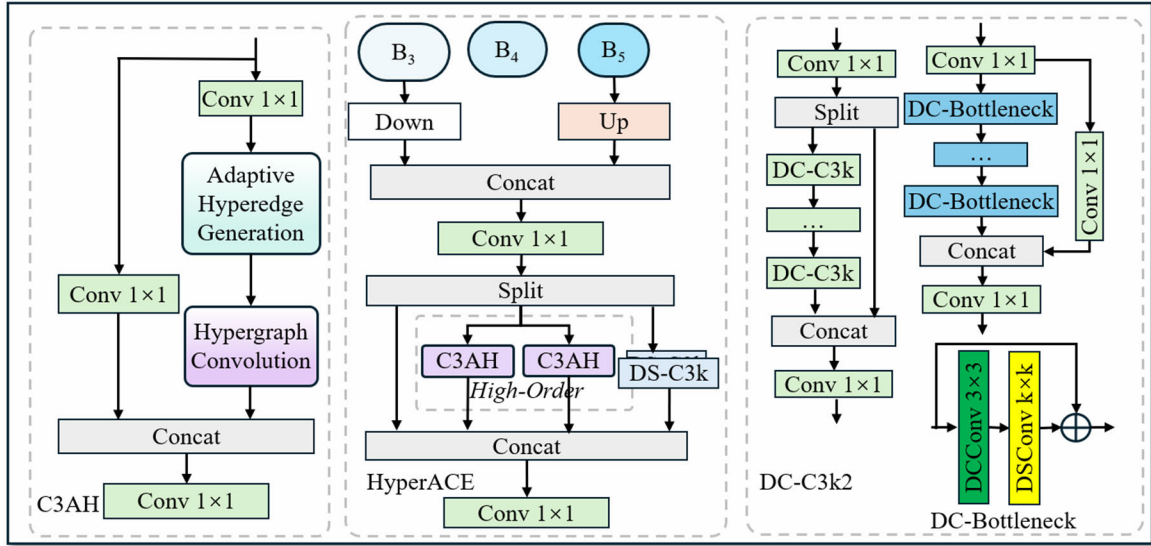


FIGURE 2. Some modules architecture.

B. YOLOv13 BASELINE

YOLOv13 is adopted as the baseline detector in this study. Its main components include Hypergraph-based Adaptive Correlation Enhancement (HyperACE) and Full-Pipeline Aggregation-and-Distribution (FullPAD). HyperACE models cross-scale and cross-position feature correlations, whereas FullPAD distributes the enhanced representations to different stages of the backbone, neck, and prediction head.

Given the multi-scale backbone features B_3 , B_4 and B_5 , HyperACE first aligns their spatial resolutions through downsampling and upsampling and then performs channel adaptation and feature fusion:

$$X_a = \text{Conv}_{1 \times 1} [\text{Concat}(\text{Down}(B_3), B_4, \text{Up}(B_5))] \quad (1)$$

The aligned feature X_a is divided along the channel dimension into a high-order branch X_h , a low-order branch X_l , and a shortcut branch X_s :

$$[X_h, X_l, X_s] = \text{Split}(X_a) \quad (2)$$

The high-order branch employs K parallel C3AH blocks to model adaptive high-order correlations, while the low-order branch uses stacked DS-C3k blocks to preserve local structural details:

$$\begin{aligned} \hat{X}_h &= \text{Concat}(\text{C3AH}_1(X_h), \dots, \text{C3AH}_K(X_h)) \\ \hat{X}_l &= \text{DSOC3k}^{(n)}(X_l) \end{aligned} \quad (3)$$

The outputs of the three branches are concatenated and fused through a 1×1 convolution to obtain the correlation-enhanced representation:

$$Y = \text{Conv}_{1 \times 1} [\text{Concat}(\hat{X}_h, \hat{X}_l, X_s)] \quad (4)$$

FullPAD subsequently adapts Y to different spatial resolutions and channel dimensions and injects the resulting

features into key stages of the detection pipeline:

$$\begin{aligned} H_i &= \text{Conv}_{1 \times 1} [\text{Resize}(Y, s_i)] \\ \hat{F}_i &= \text{Fuse}(F_i, H_i), \quad i \in \{3, 4, 5\} \end{aligned} \quad (5)$$

where s_i denotes the target spatial resolution, F_i denotes the original feature at the corresponding stage, and $\hat{F}_i$ is the feature enhanced by the FullPAD tunnel. Through HyperACE-based correlation modeling and FullPAD-based feature coordination, YOLOv13 provides a suitable baseline for the subsequent integration of IIM, BRA, and deformable convolution.

C. BI-LEVEL ROUTING ATTENTION

Under complex underwater imaging conditions such as non-uniform illumination and turbidity-induced scattering, local textures and boundaries often exhibit a coexistence of “strong noise and weak structure.” On the one hand, background highlights, specular reflections, and suspended particles may generate strong responses from pseudo-textures; on the other hand, the true edges and critical structural cues of the target are weakened, such that methods relying solely on local convolution or dense global attention are easily disturbed by unreliable regions. To improve the reliability of contextual discrimination in complex backgrounds, Bi-level Routing Attention (BRA) [29] is incorporated into the high-level feature-fusion stage of YOLOv13. It first selects, at the region level, a small number of regions most relevant to the current query, and then performs token-level attention only within these candidate regions, thereby suppressing noise propagation, strengthening effective context aggregation, and reducing computational cost simultaneously.

As shown in the figure above, Fig. 3 illustrates the principle of the Bi-level Routing Attention algorithm. Let the input feature be $X \in R^{H \times W \times C}$, It is first partitioned

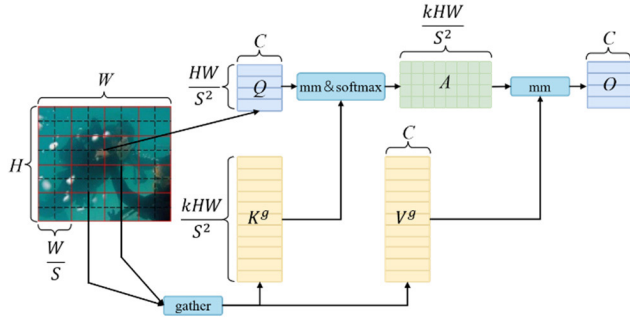


FIGURE 3. Bi-level routing attention principle.

into a set of local windows (or regions), and the corresponding Query and Key representations at the region level, Q_r and K_r , are then obtained. BRA constructs a routing graph based on region-level similarity measurement, and for each query region, selects the Top-k most relevant key regions to form the routing set:

$$\mathcal{R} = \text{TopK}(\text{Sim}(Q_r, K_r)) \quad (6)$$

Here, $\mathcal{R}$ denotes the index set of the selected candidate regions. The key idea of this step is that the sparsity pattern is not fixed, but dynamically determined by content relevance; therefore, under complex illumination and background variations, the method can preferentially retain the contextual regions that contribute most to discrimination. After obtaining the routing set $\mathcal{R}$, BRA gathers the corresponding K and V from these regions to form a candidate set, and then performs attention aggregation at the token level:

$$\text{Attn}(Q, K, V) = \text{Softmax}\left(\frac{QK_{\mathcal{R}}^T}{\sqrt{d}}\right)V_{\mathcal{R}} \quad (7)$$

D. ILLUMINATION-INVARIANT FEATURE EXTRACTION

In underwater scenes with non-uniform illumination, local overexposure, underexposure, and shadows can significantly alter the brightness and contrast statistics of the image, making the detection network more likely to rely on unstable cues such as background highlights or pseudo-textures. To suppress the contamination of illumination variation on feature representation, an Illumination Invariant Module (IIM) [30] is incorporated before the backbone, which explicitly constructs and learns illumination-invariant features and then fuses them with the original RGB information before feeding them into the detector, thereby improving discriminative reliability in low-light and low-contrast scenarios. As shown in Fig. 4, the schematic diagram illustrates the application of the IIM module within the network.

The theoretical basis of IIM originates from the Lambertian imaging model. Under the body-reflection assumption of the dichromatic reflection model, the observation of pixel p_i in the color channel $C = \{R, G, B\}$ can be written as

$$C_{p_i} = m(\vec{n}_{p_i}, \vec{l}_{p_i})e^{C_{p_i}}(\lambda)\rho^{C_{p_i}}(\lambda) \quad (8)$$

where $m(\cdot)$ is determined by the surface normal and the illumination direction and depends only on the spatial position; $e^{C_{p_i}}(\lambda)$ denotes the spectral power distribution of the illumination; and $\rho^{C_{p_i}}(\lambda)$ represents the object reflectance, which is an intrinsic property. This decomposition indicates that if both the position-related term $m(\cdot)$ by exploiting cross-channel differencing at the same spatial location, and then removes the illumination spectral term $e(\cdot)$ through inter-pixel differencing/ratio operations under the assumption that neighboring pixels share approximately consistent illumination. Taking adjacent pixels p_1 and p_2 , and the red and blue channels as an example, the Cross-Color Ratio can be defined as follows.

$$M_{rb} = \frac{R_{p_1}B_{p_2}}{R_{p_2}B_{p_1}} \quad (9)$$

Under the assumption that neighboring pixels are illuminated approximately consistently, i.e., $e^{C_{p_1}} \approx e^{C_{p_2}}$, taking the logarithm of Eq. (8) and substituting it into Eq. (9) yields a simplified form that is independent of the illumination term:

$$\log(M_{rb}) = \log(\rho^{R_{p_1}}) - \log(\rho^{R_{p_2}}) + \log(\rho^{B_{p_2}}) - \log(\rho^{B_{p_1}}) \quad (10)$$

As can be seen from Eq. (10), intra-channel subtraction can cancel the illumination term, which is later implemented through a zero-mean constraint, whereas cross-channel subtraction can eliminate the effect introduced by the position-related surface-normal/illumination-direction term. This provides the theoretical foundation for constructing learnable illumination-invariant features. Although Eq. (10) can be used to directly compute illumination-invariant features, its fixed formulation is difficult to adapt to different data distributions and downstream task requirements. To address this issue, the proposed method generalizes it into a learnable convolutional form by learning a set of $k \times k$ convolution kernels, $W_1, \dots, W_n$, and replacing the fixed differential coefficients of ± 1 with trainable weights w_i , thereby obtaining task-driven illumination-invariant representations. Taking a local neighborhood pixel set $\{p_i\}$ as an example, the extended form can be written as follows

$$M_{rb} = \prod_{i=1}^{k^2} \left(\frac{R_{p_i}}{B_{p_i}}\right)^{w_i} \quad (11)$$

To ensure that the extended formulation still satisfies illumination invariance, a zero-mean constraint is imposed on the convolution kernels so that the weighted sum of the illumination spectral terms within the same channel equals zero:

$$\begin{cases} \sum_{i=1}^{k^2} w_i \log(e^{R_{p_i}}(\lambda)) = 0 \\ \sum_{i=1}^{k^2} w_i \log(e^{B_{p_i}}(\lambda)) = 0 \end{cases} \quad (12)$$

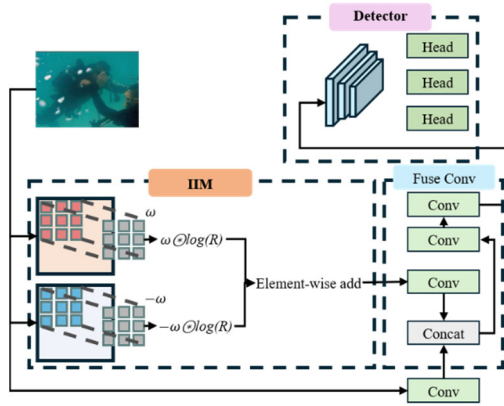


FIGURE 4. Illumination invariant module principle.

where w_i denotes the learnable coefficient associated with the i -th spatial position of the kk convolution kernel, with $i = 1, \dots, k^2$. To maintain the zero-mean constraint during training, the unconstrained kernel coefficients $\tilde{w}_i$ are reparameterized as

$$w_i = \tilde{w}_i - \frac{1}{k^2} \sum_{j=1}^{k^2} \tilde{w}_j \quad (13)$$

such that $\sum_{i=1}^{k^2} w_i = 0$. This operation is independently applied to each convolution kernel, allowing the kernel to learn task-relevant local correlations while preserving its ability to suppress illumination variations.

This constraint endows the convolution kernels with an inherent ability to suppress illumination effects at initialization, while allowing them to learn richer task-relevant correlation patterns during subsequent training without sacrificing illumination invariance. The illumination-invariant features produced by IIM are not intended to replace the original image information; instead, they are fused with RGB features before being fed into the detector, so as to jointly preserve “texture stability” and “original semantic information.” In the proposed framework, Fuse-Convolution is adopted to integrate IIM features with the original image features, and the fused representation is then passed to the subsequent detection modules. As a front-end feature enhancement branch of YOLOv13, IIM is used to generate F_{inv} before the features enter the backbone, and this representation is concatenated with the main features, thereby providing a more stable low-level representation for subsequent correlation modeling and localization regression under non-uniform illumination conditions.

E. DEFORMABLE CONVOLUTION

Under underwater conditions with non-uniform illumination and turbidity-induced scattering, target boundaries and critical structures often exhibit local missingness, blur, and spatial drift. In scenarios involving target occlusion or overlap,

the distribution of effective structural cues becomes even more incomplete, causing standard convolutions, which aggregate features over fixed regular sampling locations, to easily capture large amounts of background or noise responses, thereby leading to localization jitter and bounding-box offset during regression. To improve structural alignment and localization robustness in complex underwater scenes, deformable convolution [31] is incorporated into the key prediction layers of YOLOv13 to dynamically align feature sampling with effective target structures through adaptive sampling and spatial aggregation. Fig. 5 shows a comparison between deformable convolution and several representative core operators, including global attention, local attention, and large-kernel convolution.

DCN extends the regular sampling offsets of convolution to learnable dynamic offsets, and further combines them with a grouping mechanism to achieve more efficient adaptive spatial aggregation. Its output can be expressed as follows (taking G groups as an example):

$$y(p) = \sum_{g=1}^G \sum_{k=1}^{K^2} w_{g,k} x(p + p_k + \Delta p_{g,k}(p)) \cdot \Delta m_{g,k}(p) \quad (14)$$

where $\Delta p_{g,k}(p)$ denotes the offset of the k -th sampling point in group g at spatial position p , and $\Delta m_{g,k}(p)$ denotes the corresponding modulation coefficient. Both the sampling offsets and modulation coefficients are dynamically predicted from the input feature through a separable convolution consisting of a 3×3 depth-wise convolution followed by a linear projection. The predicted modulation coefficients are normalized using the softmax function over the K sampling points within each group, such that

$$\sum_{k=1}^K \Delta m_{g,k}(p) = 1 \quad (15)$$

Therefore, the modulation coefficients are conditioned on the input features and learned jointly with the network through backpropagation rather than being manually specified.

III. RESULTS AND DISCUSSION

A. EXPERIMENTAL DATA

The experiment on underwater target feature extraction and detection is conducted on the UT dataset [32], which is publicly available at <https://github.com/Robotics-Institute-HIST/Dataset/tree/master>. The UT dataset comprises a total of 2616 images, including 864 images of Underwater detection equipment, 304 images of Torpedo, 1009 images of Submarine, and 439 images of Frogman. The dataset is divided into a training set of 1831 images and a test set of 785 images in a 7:3 ratio.

The URPC2020(Underwater Robot Professional Contest 2020) dataset [33] encompasses a diverse range of underwater organisms, including starfish, echinus, scallop,

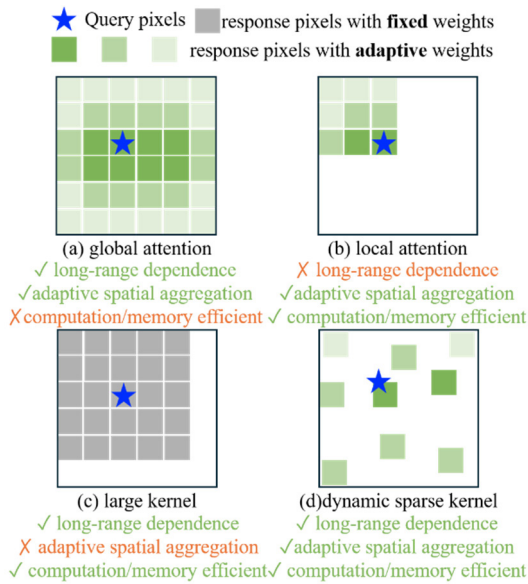


FIGURE 5. Comparisons of different core operators. (a) global attention, (b) local attention, (c) large-kernel convolution, and (d) dynamic sparse kernel. [31].

holothurian, and seagrass. After performing relevant processing on this dataset, removing interfering samples and mislabeled ones, a total of 7,350 images. The dataset comprises four types of underwater organisms: starfish, holothurian, echinus, and scallop. The number of each category is 1785, 1210, 5785, and 2757 respectively. The dataset is divided into a training set of 5145 images and a test set of 2205 images in a 7:3 ratio.

B. EXPERIMENTAL ENVIRONMENT AND HYPERPARAMETERS

The experiment was configured with 200 training iterations, an input image resolution of 640×640 , and the SGD (Stochastic Gradient Descent) algorithm to optimize the loss function. A custom learning rate decay strategy was employed during training: the learning rate was set to 0.001 for the first 40 training iterations, to 0.0001 for

iterations 40 through 80, and to 0.00001 for the final 20 iterations. The network was trained and tested on Windows 11 using the PyTorch framework, with the following hardware specifications: NVIDIA GeForce RTX 4090.

C. EXPERIMENTAL EVALUATION CRITERIA

When evaluating the output of a detection network, the predicted bounding boxes must be compared with the ground-truth bounding boxes. The Intersection Over Union (IOU) between the predicted and ground-truth bounding boxes is used to determine whether the predicted boxes match the ground-truth boxes. Based on the matching results, the detection results are classified into True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).

Precision represents the proportion of correctly detected samples out of the total number of detected samples. It is calculated as the ratio of the number of correctly predicted samples to the total number of predicted samples, and its formula is:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (16)$$

Recall represents the proportion of correctly detected samples out of the total number of true samples. It is calculated by dividing the number of correctly predicted samples in the detection results by the total number of true samples. The formula is:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (17)$$

To comprehensively evaluate the network's performance, we combine precision and recall to plot a precision-recall (P-R) curve with precision on the x-axis and recall on the y-axis. We calculate the area under the P-R curve for each category as the Average Precision (AP), and use the mean of these category-specific Average Precision values—the Mean Average Precision (mAP)—to represent the network's performance.

$$\text{mAP} = \frac{1}{C} \sum_{i=1}^C AP_i \quad (18)$$

TABLE 1. Results of the underwater target detection experiment.

Method	Underwater detection equipment(AP/%)	Submarine (AP/%)	Frogman (AP/%)	Torpedo (AP/%)	mAP/%	FPS	GFLOPs
UDTRNet	91.5	89.2	84.8	85.9	87.85	92	16.8
DMFFM	91.8	89.5	85.6	86.8	88.43	90	15.3
EHDC	91.9	90.0	88.1	86.2	89.06	94	12.1
YOLOv11	93.6	92.4	91.6	86.5	91.02	151	6.5
CSAF-YOLO	93.7	92.9	92.0	87.3	91.47	130	8.7
YOLOv12	93.5	94.5	93.9	89.1	92.76	136	6.5
YOLOv13	94.0	94.3	93.9	89.5	92.93	129	6.4
Ours	97.0	96.0	94.4	93.6	95.24	115	8.9

Here, $p(r)$ denotes the P-R curve, and C represents the number of target classes.

Additionally, the speed of each detection method is compared based on the number of frames per second (FPS).

D. EXPERIMENTAL RESULTS AND ANALYSIS

To demonstrate the effectiveness of our method, we compare it with existing detection methods: EHDC [34], DMFFM (Dynamic Multiscale Feature Fusion Method) [35], and UDTRNet [36], as well as the currently representative detection algorithms YOLOv11 [37], CSAF-YOLO [38], YOLOv12 [39], and YOLOv13 [28]—a total of seven methods. The AP, mAP, and FPS for each method across all categories are shown in Table 1. Based on this table, the comparative analysis of the results is as follows.

UDTRNet fails to effectively extract and fuse spatial and semantic information at different scales in the images, resulting in the loss of relevant spatial features. Consequently, it cannot effectively detect underwater targets where certain spatial features are obscured due to underwater turbidity and image noise, achieving an mAP of only 87.85%. DMFFM extracts four layers of multi-scale features from images using a multi-scale InfoNCE loss for fusion, achieving better detection performance than UDTRNet. EHDC expands the convolutional receptive field through dilated convolutions, enabling a fine-grained representation of small-scale target features, while simultaneously utilizing an adaptive correlation matrix to extract spatial semantic information, yielding results superior to the DMFFM.

By introducing a parallel dual-branch architecture and a spatial attention mechanism, YOLOv11 dynamically enhances key regions in feature maps, effectively improving robustness against occluded objects or complex backgrounds; its detection results outperform those of EHDC. CSAF-YOLO combines a spatial attention mechanism with a channel attention mechanism, resulting in even better detection performance. YOLOv12 builds its detection network around an attention mechanism, introducing 7×7 deep separable convolutions to help regional attention perceive positional information, thereby improving the accuracy of feature extraction and achieving better detection performance than CSAF-YOLO. YOLOv13 employs the HyperACE mechanism, which leverages hypergraph learning to capture higher-order feature correlations, enabling it to capture global semantic information across spatial positions. Additionally, by adopting a full-pipeline aggregation and distribution paradigm, it effectively improves gradient propagation, resulting in performance superior to that of YOLOv12.

In terms of computational complexity, the proposed method requires 8.9 GFLOPs, which is moderately higher than the 6.4 GFLOPs of YOLOv13 because of the additional IIM, BRA, and DCN components, but remains lower than those of UDTRNet, DMFFM, and EHDC, indicating a reasonable trade-off between detection accuracy and computational cost.

Our method outperforms other approaches, such as YOLOv13, in terms of mAP scores. This is due to the following reasons:

- 1) In underwater scenes with non-uniform illumination, detection networks are prone to relying on unstable cues such as bright background highlights or spurious textures. By introducing the IIM module, which fuses illumination-invariant features with raw RGB information through learning, we enhance the reliability of classification in non-uniformly lit scenes and suppress the noise caused by illumination variations at the feature level.
- 2) Due to strong pseudo-textures generated by underwater turbidity, scattering, and suspended particles, relying solely on local convolutions is prone to interference. The introduction of the BRA module dynamically determines and enhances the contextual regions that contribute most to classification within the global context, thereby improving the reliability of contextual information-based classification.
- 3) Underwater object boundaries often exhibit local missing or overlapping occlusion, making conventional convolutions susceptible to noise. We extend part of the conventional convolutions to a Deformable Convolution (DCN), enabling dynamic convolution shifts. This assigns higher weights to reliable structures and achieves dynamic alignment of effective sampling positions.

Observing the FPS of each method, our approach achieves an FPS of 115 while maintaining good detection accuracy, placing it among the top performers compared to other methods and meeting the requirements of engineering applications.

To analyze the advantages of our method in terms of training efficiency, we plotted an iterative accuracy curve with the mAP values of the test set for the eight methods on the y-axis and the number of training iterations on the x-axis, as shown in Figure 6.

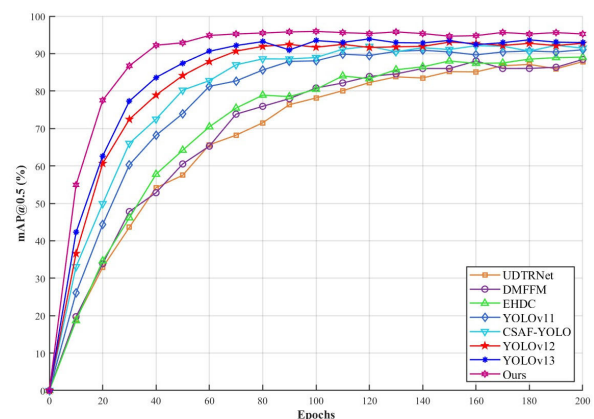


FIGURE 6. Iterative accuracy of each method.

As shown in Fig. 6, our detection accuracy has stabilized by the 80th training iteration, demonstrating a clear advantage in training efficiency compared to the baseline methods. Additionally, the overall training process of Ours is smoother, indicating superior training performance.

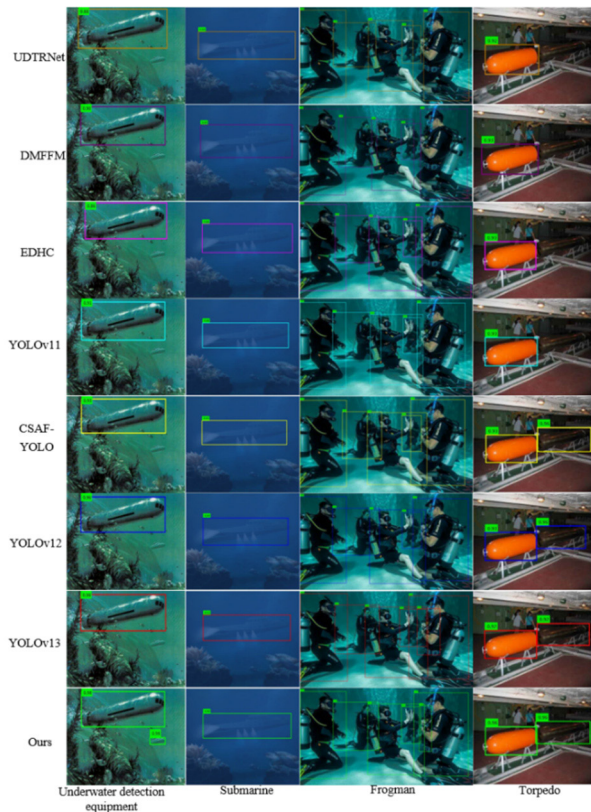


FIGURE 7. Test results for each method.

To analyze the differences in detection results across various methods, the results from the test set were visualized, as shown in Figure 7. Each row represents the detection results for a specific method, and each column represents a detection category: from left to right, these are Underwater Detection Equipment, Submarine, Frogman, and Torpedo. The boxes in the figure indicate the model's detection and localization results, and the numbers next to them represent the classification probabilities.

As shown in the Figure 7, in the 'Underwater Detection Equipment' class, only our method detected the underwater detection equipment target—which was small, blurry, and had indistinct object boundaries—while all other methods failed to detect it. In the 'Submarine' class, all methods detected the target, but there were slight differences in localization accuracy; the last five methods were more accurate and better aligned with the ground truth bounding boxes. In the 'Frogman' category, only YOLOv13 and our method detected the frogman whose figure was largely obscured by the frogman on the far right. YOLOv13's bounding box was larger than the ground truth, while our method's box was closer in size to the ground truth, resulting in higher

TABLE 2. Experimental results for different methods.

IIM	BRA	DCN	Map/%	Improvement rate/%
×	×	×	92.93	-
✓	×	×	94.31	1.38
×	✓	×	93.56	0.63
×	×	✓	94.17	1.24
✓	✓	×	94.78	1.85
✓	✓	✓	95.24	2.31

detection and localization accuracy; In the 'torpedo' category, all methods detected the torpedo on the left. Only the last four methods detected the torpedo on the right, which was difficult to detect due to uneven lighting; moreover, YOLOv13 and our method achieved higher detection and localization accuracy.

In the four representative scenes shown in Fig. 7, the proposed method detected all displayed targets and generally produced higher confidence scores and bounding boxes that were more closely aligned with the ground truth. These qualitative results suggest that the coordinated use of IIM, BRA, and DCN helps reduce the effects of illumination variation, boundary blurring, and target occlusion, thereby contributing to more stable detection and localization in complex underwater scenes.

E. ABLATION EXPERIMENT RESULTS AND ANALYSIS

To evaluate the contributions of IIM, BRA, and DCN to the YOLOv13 framework, ablation experiments were conducted, as shown in Table 2. Compared with the YOLOv13 baseline, which achieves an mAP of 92.93%, introducing IIM, BRA, and DCN individually increases the mAP to 94.31%, 93.56%, and 94.17%, corresponding to improvements of 1.38, 0.63, and 1.24 percentage points, respectively. IIM improves feature stability under non-uniform illumination by fusing illumination-invariant features with the original RGB information. BRA strengthens the aggregation of reliable contextual information, while replacing selected depthwise separable convolutions with DCN improves adaptive sampling and structural alignment. Combining IIM and BRA increases the mAP to 94.78%, yielding an improvement of 1.85 percentage points. When all three components are integrated, the mAP reaches 95.24%, representing an overall improvement of 2.31 percentage points. These results indicate that IIM, BRA, and DCN provide complementary contributions to the proposed framework.

F. COMPARISON ON THE URPC2020 DATASET

To further evaluate the underwater object detection performance of the proposed method in complex scenarios, additional experiments were conducted on the URPC2020 dataset. The proposed method was compared with the detection models listed in Table 1, and the corresponding results are presented in Table 3. As shown in Table 3, the proposed method achieves an mAP of 84.91%,

TABLE 3. Detection results on the URPC2020 dataset.

Method	Starfish (AP/%)	Holothurian (AP/%)	Echinus (AP/%)	Scallop (AP/%)	mAP/%	FPS
UDTRNet	88.6	63.3	87.1	82.4	80.36	90
DMFFM	89	62.2	87.4	82.3	80.23	89
EHDC	88.8	66.0	87.5	83.9	81.55	92
YOLOv11	89.1	67.0	87.7	84.3	82.03	143
CSAF-YOLO	90.1	71.6	89.6	84.2	83.87	129
YOLOv12	91.2	68.8	90.7	85.4	84.02	130
YOLOv13	89.7	72.0	89.8	85.1	84.15	126
Ours	90.4	72.4	90.8	86.0	84.91	113

which is 0.76 percentage points higher than that of the YOLOv13 baseline. It also obtains competitive AP values across the four underwater organism categories while maintaining an inference speed of 113 FPS.

IV. CONCLUSION

This paper addresses underwater object detection under complex optical imaging conditions, focusing on two critical issues: unstable discrimination caused by non-uniform illumination and localization drift induced by occlusion and target overlap. To this end, a stage-wise collaborative detection framework is developed based on YOLOv13. Specifically, IIM is employed at the representation stage to suppress illumination-induced feature contamination, BRA is incorporated at the feature-fusion stage to enhance the aggregation of reliable contextual information, and DCN is used at the prediction stage to dynamically align feature sampling with effective target structures. Through this coordinated configuration, the proposed method jointly improves feature representation, contextual discrimination, and target localization. Experimental results show that the proposed method achieves an mAP of 95.24% and an inference speed of 115 FPS on the UT dataset, exceeding the YOLOv13 baseline by 2.31 percentage points. On the URPC2020 dataset, it achieves an mAP of 84.91% and an inference speed of 113 FPS, outperforming the YOLOv13 baseline by 0.76 percentage points. The consistent improvements across the two datasets indicate that the effectiveness of the proposed stage-wise integration is maintained under different underwater target categories and data distributions. Ablation experiments further verify the complementary effects of IIM, BRA, and DCN within the overall framework. Overall, the proposed method improves underwater object detection accuracy while retaining favorable real-time performance, providing an effective solution for underwater target perception and search under complex imaging conditions.

REFERENCES

- [1] B. Zhang, D. Ji, S. Liu, X. Zhu, and W. Xu, "Autonomous underwater vehicle navigation: A review," *Ocean Eng.*, vol. 273, Apr. 2023, Art. no. 113861.
- [2] W. Liu, Z. Xu, J. Liu, H. Chen, and Y. Liu, "Adaptive subspace signal detection and performance analysis in nonzero-mean clutter," *IEEE Trans. Signal Process.*, vol. 74, pp. 2769–2786, 2026.
- [3] X. Zhang, P. Yang, and J. Wang, "Focusing SAS nonuniform data based on chirp scaling algorithm," *J. Franklin Inst.*, vol. 363, no. 8, May 2026, Art. no. 108666.
- [4] M. Jian, N. Yang, C. Tao, H. Zhi, and H. Luo, "Underwater object detection and datasets: A survey," *Intell. Mar. Technol. Syst.*, vol. 2, no. 1, p. 9, Mar. 2024.
- [5] Y. Ning, Y.-P. Jin, Y.-D. Peng, and J. Yan, "Low-illumination underwater image enhancement based on non-uniform illumination correction and adaptive artifact elimination," *Frontiers Mar. Sci.*, vol. 10, Sep. 2023, Art. no. 1249351.
- [6] Y. Liu, Z. Qiao, Z. Ye, W. Zhou, M. Wang, Q. He, and T. Cai, "Underwater image enhancement based on optimally weighted histogram framework and improved Fick's law algorithm," *Sci. Rep.*, vol. 14, no. 1, p. 30320, Dec. 2024.
- [7] T. Li, S. Rong, W. Zhao, L. Chen, Y. Liu, H. Zhou, and B. He, "Underwater image enhancement using adaptive color restoration and dehazing," *Opt. Exp.*, vol. 30, no. 4, pp. 6216–6235, 2022.
- [8] M. Muniraj and V. Dhandapani, "Underwater image enhancement by color correction and color constancy via retinex for detail preserving," *Comput. Electr. Eng.*, vol. 100, May 2022, Art. no. 107909.
- [9] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10778–10787.
- [10] C.-Y. Wang, A. Bochkovskiy, and H.-Y.-M. Liao, "YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7464–7475.
- [11] J. Zhang, J. Zhang, K. Zhou, Y. Zhang, H. Chen, and X. Yan, "An improved YOLOv5-based underwater object-detection framework," *Sensors*, vol. 23, no. 7, p. 3693, Apr. 2023.
- [12] H. Zhou, M. Kong, H. Yuan, Y. Pan, X. Wang, R. Chen, W. Lu, R. Wang, and Q. Yang, "Real-time underwater object detection technology for complex underwater environments based on deep learning," *Ecol. Inform.*, vol. 273, Apr. 2024, Art. no. 113861.
- [13] F. Lei, F. Tang, and S. Li, "Underwater target detection algorithm based on improved YOLOv5," *J. Mar. Sci. Eng.*, vol. 10, no. 3, p. 310, Feb. 2022.
- [14] L. Zeng, B. Sun, and D. Zhu, "Underwater target detection based on faster R-CNN and adversarial occlusion network," *Eng. Appl. Artif. Intell.*, vol. 100, Apr. 2021, Art. no. 104190.
- [15] H. Yang, T. Zhou, H. Jiang, X. Yu, and S. Xu, "A lightweight underwater target detection network for forward-looking sonar images," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–13, 2024.
- [16] W. Lu, C. Lan, C. Niu, W. Liu, L. Lyu, Q. Shi, and S. Wang, "A CNN-transformer hybrid model based on CSWin transformer for UAV image object detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1211–1231, 2023.
- [17] J. Bai, H. Hu, X. Liu, S. Zhuang, and Z. Wang, "UAV image object detection based on self-attention guidance and global feature fusion," *Image Vis. Comput.*, vol. 151, Nov. 2024, Art. no. 105262.
- [18] Y. Zhou, "A YOLO-NL object detector for real-time detection," *Expert Syst. Appl.*, vol. 238, Mar. 2024, Art. no. 122256.
- [19] C. Fu, R. Liu, X. Fan, P. Chen, H. Fu, W. Yuan, M. Zhu, and Z. Luo, "Rethinking general underwater object detection: Datasets, challenges, and solutions," *Neurocomputing*, vol. 517, pp. 243–256, Jan. 2023.

- [20] Y. Hao, Y. Yuan, H. Zhang, and Z. Zhang, "Underwater optical imaging: Methods, applications and perspectives," *Remote Sens.*, vol. 16, no. 20, p. 3773, Oct. 2024.
- [21] L. Chen, Y. Huang, J. Dong, Q. Xu, S. Kwong, H. Lu, H. Lu, and C. Li, "Underwater optical object detection in the era of artificial intelligence: Current, challenge, and future," *ACM Comput. Surveys*, vol. 58, no. 3, pp. 1–34, Feb. 2026.
- [22] L. Dai, H. Liu, P. Song, and M. Liu, "A gated cross-domain collaborative network for underwater object detection," *Pattern Recognit.*, vol. 149, May 2024, Art. no. 110222.
- [23] Q. Cen, Q. Zhu, Y. Wang, W. Chen, and S. Liu, "YOLOv9-YX: Lightweight algorithm for underwater target detection," *Vis. Comput.*, vol. 41, no. 6, pp. 4033–4045, Apr. 2025.
- [24] J. Gao, Y. Zhang, X. Geng, H. Tang, and U. A. Bhatti, "PE-transformer: Path enhanced transformer for improving underwater object detection," *Expert Syst. Appl.*, vol. 246, Jul. 2024, Art. no. 123253.
- [25] L. Chen, F. Zhou, S. Wang, J. Dong, N. Li, H. Ma, X. Wang, and H. Zhou, "SWIPENET: Object detection in noisy underwater scenes," *Pattern Recognit.*, vol. 132, Dec. 2022, Art. no. 108926.
- [26] B. Wang, Z. Wang, W. Guo, and Y. Wang, "A dual-branch joint learning network for underwater object detection," *Knowl.-Based Syst.*, vol. 293, Jun. 2024, Art. no. 111672.
- [27] C. Li, W. Liu, G. Gong, X. Ding, and X. Zhong, "SU-YOLO: Spiking neural network for efficient underwater object detection," *Neurocomputing*, vol. 644, Sep. 2025, Art. no. 130310.
- [28] M. Lei, S. Li, Y. Wu, H. Hu, Y. Zhou, X. Zheng, G. Ding, S. Du, Z. Wu, and Y. Gao, "YOLOv13: Real-time object detection with hypergraph-enhanced adaptive visual perception," 2025, *arXiv:2506.17733*.
- [29] L. Zhu, X. Wang, Z. Ke, W. Zhang, and R. Lau, "BiFormer: Vision transformer with bi-level routing attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Vancouver, BC, Canada: IEEE, Jun. 2023, pp. 10323–10333.
- [30] M. Hong, S. Cheng, H. Huang, H. Fan, and S. Liu, "You only look around: Learning illumination-invariant feature for low-light object detection," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 37, 2024, pp. 87136–87158.
- [31] W. Wang, J. Dai, Z. Chen, Z. Huang, Z. Li, X. Zhu, X. Hu, T. Lu, L. Lu, H. Li, X. Wang, and Y. Qiao, "InternImage: Exploring large-scale vision foundation models with deformable convolutions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 14408–14419.
- [32] Q. Sun and L. Cai, "Multi-AUV target recognition method based on GAN-meta learning," in *Proc. 5th Int. Conf. Adv. Robot. Mechatronics (ICARM)*, Dec. 2020, pp. 374–379.
- [33] C. Liu, H. Li, S. Wang, M. Zhu, D. Wang, X. Fan, and Z. Wang, "A dataset and benchmark of underwater object detection for robot picking," in *Proc. IEEE Int. Conf. Multimedia Expo Workshops (ICMEW)*, Jul. 2021, pp. 1–6.
- [34] L. Cai, X. Qin, and T. Xu, "EHDC: Enhanced dilated convolution framework for underwater blurred target recognition," *Robotica*, vol. 41, no. 3, pp. 900–911, Mar. 2023.
- [35] L. Cai, Y. Li, C. Chen, and H. Chai, "Dynamic multiscale feature fusion method for underwater target recognition," *J. Sensors*, vol. 2022, pp. 1–10, Jul. 2022.
- [36] L. Cai, C. Chen, and H. Chai, "Underwater distortion target recognition network (UDTRNet) via enhanced image features," *Comput. Intell. Neurosci.*, vol. 2021, no. 1, Jan. 2021, Art. no. 4193625.
- [37] R. Khanam and M. Hussain, "YOLOv11: An overview of the key architectural enhancements," 2024, *arXiv:2410.17725*.
- [38] H. R. Zhang, W. M. Feng, and L. X. Yang, "CSAF-YOLO: Improved YOLO11 algorithm for underwater small object detection," *J. Comput. Appl.*, vol. 4, no. 2, pp. 1–12, 2026.
- [39] Y. Tian, Q. Ye, and D. Doermann, "YOLOv12: Attention-centric real-time object detectors," 2025, *arXiv:2502.12524*.



GAOQUAN GU received the Ph.D. degree in engineering from the Naval University of Engineering. He is currently with the Naval University of Engineering. He has more than ten years of experience in the field of intelligent sensors. He has led and completed three provincial-level scientific research projects and participated in one project funded by the National Natural Science Foundation of China. He has published more than 12 papers and has been granted seven patents. His research interests include acoustic image processing and artificial intelligence algorithms.



KAI LIU received the Ph.D. degree from the Naval University of Engineering. He has long been engaged in research on underwater detection. He has proposed a double-shell design method for underwater deep-diving vehicles, completed more than 20 provincial and ministerial research projects, received over ten authorized patents, and won one First Prize of Provincial and Ministerial Science and Technology Progress Award.



QINGCHAO YANG received the Ph.D. degree from the Naval University of Engineering. He has long been engaged in research on vibration and noise control technologies for marine power equipment as well as its maintenance support technologies. He has proposed dynamic analysis methods for multiple types of quasi-zero stiffness vibration isolation systems, completed more than ten scientific research projects including those funded by the National Natural Science Foundation of China, published over 20 high-level academic papers, and has been granted more than ten patents.

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